

**Nationality**  
French  
**Current home**  
Oxford, UK  
**Website**  
Personal [✉](#)

# Edith Darin

Demography, Bayesian statistics and Geospatial data science

## Contact

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## Tech repo

**Github**  
edarin  
**OpenStreetMap**  
edithdarin

## Programming

**OS Preferences**  
Windows  
GNU/Linux  
**Statistical Programming**  
R (+++)  
Python (++)  
**Dashboard & API**  
Docker, Flask,  
PostgreSQL, DuckDB,  
R Shiny, Git, Nginx  
**GIS & Remote Sensing**  
QGIS, Google Earth  
Engine, GDAL,  
PostGIS

## Languages

**French** native  
**English** fluent  
**German** intermediate  
**Italian** intermediate

## Work

- 2026 - **Population Data Scientist** Demographic Unit, University of Oxford, UK  
2024 - *-Secratio project:* technical lead for data infrastructures linking survey data to daily environmental stressors (temperature, air pollution, conflict events). *Methods developed:* Duckdb interface for Parquet files. PI: Dr Joshua Wilde  
*-NowPop project:* Model monthly forced internal displacement in war-struck Ukraine. *Methods developed:* Statistic modelling of flows of mobile phone subscribers. Automated dockerised data pipeline. Deploy nginx dashboard and flask API for dissemination. Partner: WHO. PI: Dr Douglas Leasure
- 2024 - **Research Fellow** WorldPop, University of Southampton, UK  
2018 - *Until 08/21 Specialist Research Technician*  
Lead modeller for filling population gaps in censuses impacted by insecurity (Burkina Faso, Mali, Demographic Republic of Congo). Map refugee population in Cameroon. Partners: UNFPA, UNHCR, Gates foundation, CIESIN. *Methods developed:* Hierarchical Bayesian modelling combining survey data with geospatial covariates. Random-forest models for producing high-resolution population estimates. Grid-based spatial sampling design for household surveys.
- 2017 **Data Scientist Intern** mVAM, UN World Food Program, Italy  
Assessing food security from remote sensing and mobile data collection. Machine-learning based propensity score weighting in complex survey design. Support mission for Mali Country Office (Bamako).
- 2016 **Data Scientist Intern** Etalab, Governmental Task Force for OpenData, France  
*OpenFisca:* French tax and benefit system micro-simulation platform. Finalist of the Open Government Partnership Hackathon.

## Grants

- 2021 **Mapping refugees with high spatial resolution** UNHCR: \$80 000  
Co-I for developing methods to map refugees for survey purposes.
- 2022 **Statistical population modelling for census support** UNFPA: \$30 000  
Co-I for creating a population model with Brazil Statistics Office (IBGE).

## Education

- 2026 - **DPhil in Demography** University of Oxford, UK  
2022 *Topic: Estimating subnational population sizes in data-scarce contexts.* Affiliated with the Max Planck Institute for Demographic Research and in collaboration with the Colombian Stats Office
- 2019 - **MSc in Applied GIS and Remote Sensing** University of Southampton, UK  
2018 Dissertation featured in a peer-reviewed article.
- 2018 - **MSc in Statistics and Economics** ENSAE, France  
2015 Semesters abroad at Humboldt University and Galatasaray University.

# Academic outcomes

## Peer-reviewed papers

- Leveraging high-frequency digital data to analyze forced displacement dynamics: A case study from the Gaza Strip. **Darin E**, Kashyap, R. and Leasure, D.R. *Population and Development Review* 12(4), 2026. [↗](#)
- Developing a customized, enumeration area-based sampling frame tailored to a specific population subgroup using geospatial methods, Qader S., **Darin E.**, Dicko A.H., Galal H., Park H., Jimenez R.M., Harfoot A., and Tatem A.J. *Journal of Survey Statistics and Methodology* 14(1), 2026. [↗](#)
- Demographic figures at risk in the digital era: Resisting commodification, reclaiming the common good. **Darin E.** *Big Data & Society* 12(4), 2025. [↗](#)
- How accurate are high-resolution settlement maps at predicting population counts in data-scarce settings?, **Darin E.**, Kashyap, R. and Leasure, D.R. *Population, Space and Place*, 31: e70083, 2025. [↗](#)
- Tackling public health data gaps through Bayesian high-resolution population estimation: A case study of Kasai-Oriental, Democratic Republic of the Congo, Boo G, **Darin E**, Chamberlain HR, Hosner R, Akilimali PK, et al. 2025 *PLOS Global Public Health*, 5(9): e0005072. [↗](#)
- Mapping refugee populations at high-resolution by unlocking humanitarian administrative data, **Darin, E.**; Dicko D. H.; Galal H.; Jimenez R.M.; Park H.; Tatem A.J.; Qader S., *Journal of International Humanitarian Action*, 2024 [↗](#)
- Building footprint data for countries in Africa: To what extent are existing data products comparable?, Chamberlain H. R.; **Darin, E.**; Adewole A. W.; Lazar A.; Tatem A.J., *Computers, Environment and Urban Systems*, 2024, 102104, [↗](#)
- The Population Seen from Space: When Satellite Images Come to the Rescue of the Census, **Darin, E.**; Kuépié M.; Bassinga H, Boo G., Tatem A.J.; *Population*, 2022, 77:3, [↗](#)
- High-resolution population estimation using household survey data and building footprints, Boo, G; **Darin, E.**; Leasure, D.R.; Dooley, C.A.; Chamberlain, H.R., et al, *Nature Communications*, 2022, 13:1, [↗](#)
- The role of earth observation in an integrated deprived area mapping "system" for low-to-middle income countries, Kuffer, M.; Thomson, D.R.; Boo, G.; Mahabir, R.; Grippa, T.; Vanhuysse, S.; Engstrom, R.; Ndugwa, R.; Makau, J.; **Darin, E.**; de Albuquerque, J.P. and Kabaria, C, *Remote Sensing*, 2020, 12, 982. [↗](#)
- A grid-based sample design framework for household surveys, Boo G; **Darin E**; Thomson DR and Tatem AJ, *Gates Open Research* 2020, 4:13, [↗](#)

## Pre-prints

- Lazar, A. N.; Boo, G.; Chamberlain, H. R.; Nnanatu, C. C.; **Darin, E.**; Leasure, D. R.; Yankey, O.; Gadiaga, A.; Juran, S.; de la Rua, L.; Espey, J.; Tatem, A. J. Advances in Small Area Population Estimation in the Absence of National Census Data. Preprints 2025, 2025061625. [↗](#)

## Books chapter

- Demographic figures at risk in the digital era: Resisting commodification, reclaiming the common good **Darin, E.** in *Digital Fairness and data protection in the age of artificial intelligence*, Lefebvre Giuffre. [↗](#)

## R Package

- wopr: An R package to query the WorldPop Open Population Repository, version 1.0.1, Leasure DR; Bondarenko M; **Darin E**; Tatem AJ, *WorldPop, University of Southampton*, 2020. Maintainer: **Darin E.** [↗](#)

# Awards

- Julia Mead Knox Memorial Prize for Research in Demography, University of Oxford (1000£), 2025
- Three-year scholarship from the Leverhulme Centre for Demographic Research (3x32,000), 2022, Light Senior scholarship, St Catherine's college, 2024
- Vice-Chancellor Award for Research Impact (2018), Faculty Dean's Prize for Research (2021), University of Southampton
- Four-year scholarship from the École Normale supérieure de Cachan (4x15,600€), 2015, Scholarship Sarah Andrieux for Social and Economics Research (800€), 2017